Justin Kauff

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EDUCATION

Elon University, School of Engineering and Computing | Elon, NC — Aug 2023 - Present

B.S. Engineering, Electrical and Computer Concentration, Minor in Mathematics — GPA: 3.998

- *Honors & Accomplishments:* Six-time President's List, First-Year Physics Award
- *Relevant Coursework*: Software Design, Computer Vision, Signals and Systems, Controls and Dynamic Systems, Circuit Analysis

RESEARCH EXPERIENCE

Robotics and Intelligent Systems Engineering Research Lab (RISE) | Elon University – *Dr. Blake Hament* — Apr 2024 - Present

Engineering Research Student

A Rigid Harness with Haptic Feedback for Robotic and Traditional Service Dogs — Jan 2026 – May 2026

- Designed a real-time embedded control system on an Arduino UNO R4, integrating multiple ultrasonic sensors with independently driven DC motors via dual H-bridge drivers and PWM control.
- Developed a time-of-flight sensing pipeline and nonlinear signal-mapping algorithm to convert live sensor data into calibrated actuator output.

High-Agility Dynamics Analysis for a Limited Quadruped Platform — Apr 2024 - May 2025

- Developed high-agility behaviors for the Unitree Go1 and HiWonder PuppyPi quadrupeds using ROS, Python, and C++.
- Designed repeatable motion sequences for the quadrupeds to enable consistent data collection with motion capture systems.
- Applied Fourier analysis to motion data to identify frequency-domain characteristics and determine optimal stability parameters.

PUBLICATIONS

- A. J. Keefe, **J. Kauff**, and B. Hament, "A Rigid Harness with Haptic Feedback for Robotic and Traditional Service Dogs," *2026 Systems and Information Engineering Design Symposium (SIEDS)*, 2026, doi:10.1109/SIEDS69358.2026.11540280.
- A. J. Keefe, Z. Sherpbier, M. Dion, **J. Kauff** and B. Hament, "Exploring Dynamic Analysis of a Quadrupedal Mobile Robot Modified for Animatronic Entertainment in Musical Theatre Production," *2025 Systems and Information Engineering Design Symposium (SIEDS),* 2025, doi:10.1109/SIEDS65500.2025.11021214.

PROFESSIONAL EXPERIENCE

REK | *In Collaboration with Elon's RISE Lab* — Sep 2026 - Present

Robotics Engineer Intern

- Supporting REK's development of combat-ready humanoid robots by benchmarking a Unitree G1's striking performance, collecting punch-force data via a company-provided control policy and a load-cell-based sensor.
- Integrated and tested a VR-headset teleoperation pipeline that maps operator motion to the Unitree G1 in real time.
- Fine-tuning REK's PPO-based policy with custom reinforcement learning updates to increase the G1's punch and kick strength for competitive robot combat.

Red Oak Brewery | Whitsett, NC — Sep 2026 - Present

Integration Lead

- Communicating with Red Oak Brewery to translate their operational requirements into a system design.
- Scoping a closed-loop PID control system to modulate valve position based on real-time flow-rate feedback, targeting the brewery's specified transfer rate of 2 hL/hr (±2%).
- Coordinating across mechanical, electrical, and software subsystems to meet client constraints, including retaining the brewery's existing CO_2-driven transfer process.
- Incorporating brewery staff feedback on usability and safety requirements (e.g., independent manual emergency shutoff) into the design.

PROJECTS

2-DOF Ball Balancing System | *Arduino, Embedded C, PID Control, MATLAB*

- Built a real-time closed-loop control system to balance a steel ball using two servo actuators and a resistive touchscreen sensor.
- Integrated touchscreen sensor readings with the control loop to provide real-time position feedback for actuator correction.
- Implemented PID control on Arduino using calibrated position data converted from raw ADC values to millimeter coordinates.
- Modeled and analyzed system behavior in MATLAB/Simulink to tune controller response and stability.

Vision-Based Finger-Painting System | *Python, OpenCV, YOLOv8*

- Built a real-time computer vision application that transforms webcam input into an interactive finger-painting canvas.
- Tracked fingertip position using HSV color segmentation, morphological filtering, contour detection, and image moments.
- Integrated YOLOv8 object detection to dynamically change brush color based on objects recognized in the scene.

WORK EXPERIENCE

Shames JCC on the Hudson | Tarrytown, NY — Mar 2023 – Aug 2026

Head Lifeguard

- Led a team of 40+ lifeguards, providing guidance and support to maintain a safe, welcoming environment.
- Collaborated with management to schedule staff, maintain equipment, and support smooth daily operations.

SKILLS

Programming Languages/Libraries: Java, C/C++, Python, Mathematica, ROS, MATLAB, OpenCV, NumPy, TensorFlow, Simulink

Software & Tools: Git/GitHub, Hugging Face, AWS SageMaker, SolidWorks, PyCharm, VS Code, Office 365, Google Suite, Adobe Suite

Hardware: Raspberry Pi, Arduino